

M Harshavardhana

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PROFESSIONAL SUMMARY

Computer Science undergraduate with hands-on experience in full-stack development, artificial intelligence, machine learning, and computer vision through internships and end-to-end projects. Experienced in building web applications and REST APIs using React, Node.js, Express.js, FastAPI, PostgreSQL, and Prisma, with additional experience in RAG pipelines, multi-agent systems, and deep learning. Strong foundation in Java, Python, SQL, Data Structures and Algorithms, Object-Oriented Programming, and software development.

EXPERIENCE

CodeAlpha **Sep 2026 – Oct 2026**
Full-Stack Development Intern *1 Month*

- Developed full-stack applications using React, TypeScript, TanStack Start, Node.js, Express.js, PostgreSQL, Prisma, REST APIs, and JWT authentication.
- Built ShopSphere (e-commerce), Connectly (social media), and TaskFlow (project management), implementing responsive UI, backend endpoints, database models, authentication, and CRUD workflows.

Unlox Academy **2026**
Campus Ambassador *1 Month*

- Represented Unlox Academy at Global Academy of Technology, promoting student programs and educating students about opportunities and participation.

PROJECTS

AI Agent-Based Virtual Campus Tour and Query Assistant **GitHub**

- Developed an AI-powered campus query and virtual tour system for information retrieval, interactive navigation, and voice-assisted guidance.
- Designed a supervisor-led multi-agent architecture for admissions, academics, facilities, and navigation; implemented LLM-powered RAG for Natural Language Processing (NLP) using SentenceTransformers, ChromaDB, BM25, and SVM reranking, backed by PostgreSQL/PostGIS.

Deep Learning-Based Multi-Temporal Satellite Image Change Detection System **GitHub**

- Built a deep learning pipeline to detect structural changes between before-and-after satellite image pairs using the LEVIR-CD dataset.
- Implemented Siamese U-Net for pixel-level binary change masks and added change-region quantification, building localization, and overlay visualization.

CodeAlpha Full-Stack Development Projects **GitHub**

- ShopSphere:** E-commerce application with products, cart, checkout, authentication, admin workflows, and REST API integration.
- Connectly:** Social platform with profiles, posts, likes, comments, follows, feeds, explore, and search using React/TanStack, Express, PostgreSQL, and Prisma.
- TaskFlow:** Project management tool with boards, task assignments, comments, notifications, and workflow management.

ACHIEVEMENTS

- Smart India Hackathon 2025:** Selected through the internal college hackathon for participation in SIH 2025.
- IBM Z Datathon 2025:** Participated and submitted an AI/ML project focused on climate-health risk prediction.

CERTIFICATIONS

- Front End Web Developer** — Infosys Springboard
- Cloud Computing Fundamentals** — Simplilearn

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, SQL **Frontend:** HTML5, CSS3, React, Next.js, TanStack Start, Vite, Tailwind CSS

Backend & Databases: Node.js, Express.js, FastAPI, REST APIs, JWT, MongoDB, Redis, NoSQL, PostgreSQL, Prisma, PostGIS, ChromaDB

AI/ML & Tools: Machine Learning, Deep Learning, PyTorch, Scikit-learn, RAG, AI Agents, SentenceTransformers, BM25, SVM Reranking, Siamese U-Net, Git, GitHub, Docker

EDUCATION

Global Academy of Technology, Bangalore **Aug 2023 – Jun 2027**
B.E. in Computer Science and Engineering *CGPA: 7.07*
Narayana PU College **2023**
XII (State Board) *76%*
Bunts Sangha RNS Vidyaniketan **2021**
X (ICSE) *88%*